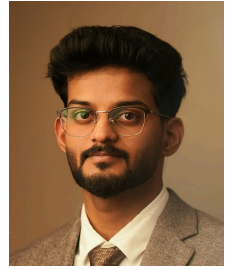


PRAVEEN VIGHNESH S T

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Websites - [Portfolio](#) | [Linkedin](#)



I'm not only skilled in CAD design and product development but also fluidly intelligent, able to learn new concepts quickly, adapt to diverse challenges, and deliver creative, well-engineered solutions across different projects.

SOFTWARE TOOLS

- | | | |
|-----------------------|--------------------|-------------|
| • Autodesk Fusion 360 | • Matlab Simulink | • Python |
| • Creo Parametric | • Ansys Mechanical | • MS Office |
| • Ansys Maxwell | • JMAG | • Power BI |

GERMAN EXPERIENCE

Factory Automation and Production Systems, Nürnberg - Master Thesis

02/2025 - present

Concepts: IOT, CAN bus, kinematics, DOF, Coordinate frame assignment, PLC, Computer Vision

- Development of a Modular Cartesian Pick and place robotic system for a smart kitchen automation unit.
- Kinematic design and control of a 5-DOF vertical gantry with rotational end-effector.
- Designing an automatic tool changer for the end effector to be able to interact with different objects inside the machine.

Factory Automation and Production Systems, Nürnberg - Project Thesis

07/2025 - 01/2025

Concepts: Model based System Engineering (MBSE), Process/Design change requests, Technical Documentation, DFM/DFA, Root Cause Analysis (RCA), BOM creation

- Design and Validation of **Mechatronic Subsystems** for an automation machine Using CAD and Strength Simulation.
- Engineered modular components with Lego-style fitment, enabling rapid part replacement and design iteration – reducing 3D printing time from **13 h to 6 h** and minimizing material waste by eliminating support structures through optimized geometry.
- Designed a grasshopper mechanism as a **speed multiplier**, increasing actuator output speed from 30 mm/s to 72 mm/s while **maintaining force requirement** of 10N for dispensing.
- Conducted **interference analysis** for dynamic parts and **tolerance analysis** for static parts across assemblies to ensure seamless motion and manufacturability.
- Created the complete **electrical circuit diagram** before procurement, enabling **efficient component sourcing** and on-schedule project completion.
- Programmed **Python-based control logic** on Raspberry Pi OS to synchronize actuator, sensor, and **motor** operations for fully automated system functionality.
- Manufactured a sheet-metal casing for a liquid storage module by laser-cutting anodized aluminum sheets, forming bends on a sheet-metal bending machine, and assembling the enclosure using rivets.

Café Mengin, Erlangen - Barista

12/2023 - 07/2025

Concepts: 5S, Inventory Management, Lean Workflow, Quality Control, Order and Queue Management

- Applied 5S and visual organization to reduce search time and stabilize workflow during rush hours.
- Supported quality consistency by standardizing preparation steps and monitoring defects.

OTHER EXPERIENCE

ISOLogic Automation, Tiruppur - Intern as Design Engineer

08/2022 - 12/2022

Concepts: CFD, FEA, Thermal analysis, Material Research, SAP modules, Machining, Welding, tolerance analysis (GD&T)

- Collaborated with cross-functional teams to design and manufacture a 41-foot Pratt truss structure, through precision cutting, welding, machining and other advanced manufacturing methodologies.

Cognizant Technology Solutions, Chennai - Programmer Analyst Trainee

02/2022 - 07/2022

Concepts: Automated Testing (Selenium, TestNG), Benchmarking, Functional and regression testing, black box/Grey box testing, Quality Assurance, Integration testing.

- Implemented automated testing with Selenium & TestNG to improve efficiency, increase test coverage and stability, and cut post-release defects by 15%.

Elite Techno Groups, Bengaluru - Intern

11/2020 - 01/2021

Concepts: Suspension and Chassis Modeling, Assembly Design Principles, Tolerance and Fit Analysis (GD&T), Steering Geometry & Axis Inclination (SAI), PDM, PLM, Vehicle Dynamics principles

- Developed strong understanding of vehicle dynamics (understeer/oversteer) and analyzed steering, suspension, chassis, and braking systems to improve handling, stability, and load response.

EDUCATION

M.Sc. Electromobility, FAU Erlangen, Germany – GPA: 2.2

10/2023 – Present

B.Tech. Mechanical Engineering, VIT, Vellore – GPA: 2

07/2018 – 06/2022

CERTIFICATES AND ACHIEVEMENTS

- Physics Hackathon – 1st Place
- AutoCAD for Design and Drafting
- Developing Innovative Ideas
- Electric Vehicles and Mobility
- Introduction to Self-Driving Cars
- Engineering Design with Fusion360

PROJECTS

Modular Electric Traction Motor Design and Strength Evaluation

Tools and Concepts: Creo Parametric, ANSYS Mechanical, Parametric Modeling

- Designed parametric CAD models of electric motor housing, stator support, and mounting fixtures, enabling rapid geometry iterations for different motor sizes.
- Performed static strength analysis on housing and fixture components under torque-induced loads, identifying stress-critical regions and improving structural robustness.
- Reduced component mass by $\approx 10\text{--}15\%$ through geometry optimization while maintaining safety factors, improving manufacturability and integration feasibility.

Thermal Behavior Analysis of an Electric Motor Stator Assembly

Tools and Concepts: ANSYS Mechanical, MATLAB, Heat Transfer Analysis

- Modeled a simplified electric motor stator assembly and analyzed heat generation and dissipation under continuous operating conditions.
- Evaluated temperature distribution across stator core and housing to identify hot-spot regions and made cooling-oriented design improvements.

Vision-Based Foot Gesture HMI for a Medical Robot

Tools and Concepts: Python, OpenCV, Numpy, VS Code, Unity, ROS2, Real-Time Computer Vision, FSM, UDP Communication

- Designed a hands-free human-machine interface that translates foot gestures into real-time control commands using computer vision.
- Implemented a robust finite state machine with temporal filtering, baseline calibration, and tracking-loss tolerance to achieve reliable gesture recognition under fast motion and partial occlusion.
- Design einer **hands-free Human-Machine Interface**, die **Foot Gestures** mit **Computer Vision** in **Real-Time Control Commands** übersetzt.
- Implementierung einer **robusten Finite State Machine**, mit **temporal filtering**, **baseline calibration** und **tracking-loss tolerance**, um eine **zuverlässige Gesture Recognition** auch bei schneller Bewegung und **partial occlusion** zu erreichen.

FPV Quadcopter Frame Design and Structural Optimization

Tools and Concepts: Fusion 360, parametric CAD Modeling, Topology Optimization, Interference Analysis, PDM with OSD

- Designed a custom X-frame quadcopter chassis to integrate motors, ESCs, PDB with OSD, camera, VTX, and LiPo battery.
- Optimized frame geometry for weight reduction and structural rigidity with mounting features for motors and electronics.

Anthropomorphic Robot Hand Design

Tools and Concepts: Fusion 360, Reconstruction from orthographic views, Joint Articulation and Assembly design

- Built a two four-bar linkage for fingers with an under-actuated mechanism which operates the MCP joint to move the PIP and DIP joint **dependently**.
- The trajectory of the joints traces a similar motion to the human hand prior to touching an object with a **single linear motor** to trigger the motion.

Adaptive Gripper with EDS-based Sensing

Tools and Concepts: Electrostatic Distance Sensing, Signal Processing, Impedance measurement

- Designed an adaptive parallel jaw gripper printed with TPU95A for flexibility and equipped with EDS foam for touch detection during compliant grasping.

Rotor Head Assembly Design & Dynamic Gyroscopic Analysis

Tools and Concepts : Dynamics & Gyroscopic Precession, Control mechanics

- Designed a rotor head assembly to investigate and visualize the principles of swashplate mechanics and gyroscopic precession relevant to helicopter and rotorcraft dynamics.

Autonomous Navigation System for Wheelchairs - Bachelor Thesis

Tools and Concepts : Solidworks, Gazebo, ROS, Python, 3D Point Cloud Localization (PCL), SLAM

- Simulated a retrofit for medical wheelchairs, Implemented intelligent mobility features including obstacle avoidance, vision-based line following, goal-directed navigation, and maze solving.

Omnidirectional AGV - Independent project

Tools and Concepts : Kinematics Simulation, CAD Assembly, Motion Control

- Designed and simulated an Omnidirectional Automated Guided Vehicle (AGV) with Mecanum wheels for 360° mobility and precise navigation in constrained environments.

Design of Boston Dynamics Spot - Independent project

Tools and Concepts : Fusion 360, Reconstruction from orthographic views, Joint Articulation and Assembly design

- Designed a high-fidelity Boston Dynamics' Spot robot model with orthographic reference drawings by creating simple and defined sketches, considering Tolerance and interactions of components.

Lockheed L-1011 TriStar Aircraft Structural Model - Independent Project

Tools and Concepts : Surface Modeling, Structural Assembly Design, Aerodynamics

- Applied parametric design and constraint-based modeling to replicate realistic joint assemblies between wing, fuselage, and tail sections.

LANGUAGES

- | | | |
|-----------------------|------------------------------|---------------------------|
| • TAMIL - C2 - Native | • GERMAN - B1 - Intermediate | • ENGLISH - C1 - Advanced |
|-----------------------|------------------------------|---------------------------|

HOBBIES

- | | | |
|---------------------------------|---------------------|----------------------------|
| • Pencil sketching and painting | • Jiu Jitsu and MMA | • Singing and Song Writing |
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